ET AI HACKATHON 2026

Scam Shield

AI-powered detection and citizen response for digital-arrest scams and impersonation fraud

Problem Statement 6: AI for Digital Public Safety — Defeating Counterfeiting, Fraud & Digital Arrest Scams

1. Problem Context

India registered 1.14 million cybercrime complaints in 2023, up 60% year-on-year, and the trend has continued to steepen. Within that category, “digital arrest” scams — where fraudsters impersonate CBI, ED, or Customs officers and hold victims in a multi-day psychological hostage situation over video call — cost Indian citizens over ₹1,776 crore in just the first nine months of 2024, per Ministry of Home Affairs reporting.

These are not opportunistic crimes. They are industrialised operations that follow a repeatable script: authority impersonation, forced isolation from family, manufactured urgency, and a fund transfer demand. Law enforcement's gap is not evidence after the fact — it is intelligence before mass victimisation occurs, and a reliable way to flag the pattern at the point of contact, not the point of complaint.

2. Solution Overview

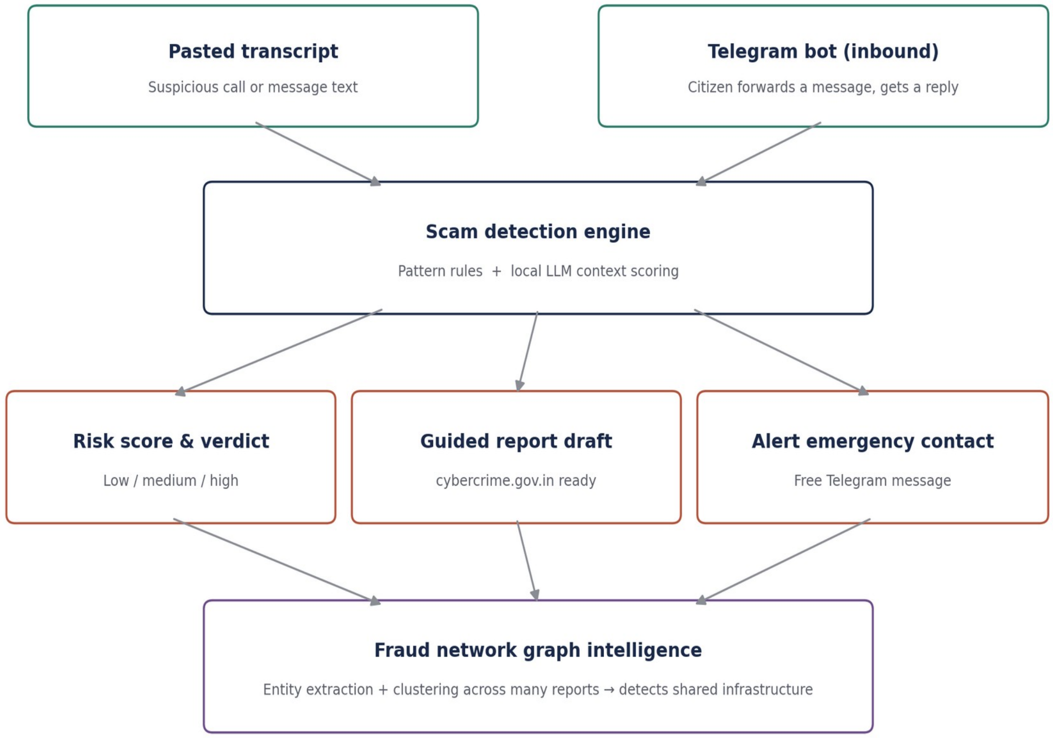
Scam Shield is a hybrid AI detection system that scores any call transcript or forwarded message for digital-arrest scam risk in real time, explains exactly which phrases triggered the score, and walks the citizen through what to do next — without needing them to first realise they are being scammed.

The system covers two pillars from the problem statement, built to production depth rather than five built shallowly:

- Digital Arrest Scam Detection & Alerting Engine — the detection brain, with a quantitative evaluation (Section 4).
- Fraud Network Graph Intelligence — cross-references multiple flagged reports to detect shared infrastructure (phone numbers, accounts, UPI IDs, case references), surfacing organised fraud rings rather than isolated incidents.

A citizen-facing channel is included via a real inbound Telegram bot (Section 3.1): citizens forward a suspicious message directly and receive an automated risk verdict — this is a working listener, not a described-but-unbuilt feature.

3. System Architecture



Two input channels feed a single detection engine: a direct paste, and a real inbound Telegram bot. The engine's three outputs (risk verdict, guided report draft, and an optional Telegram alert) also feed a fourth stage, the fraud network graph, which looks across many flagged reports rather than one at a time.

3.1 Telegram bot — inbound, not just outbound

An earlier draft of this document described a Telegram forwarding flow that only had outbound alerting built. That gap has been closed: backend/telegram_bot.py is a real listener (long polling via the official Telegram Bot API, python-telegram-bot) that receives a forwarded message and replies with a live risk verdict and next-step guidance. It runs as a separate process alongside the API server. It requires a bot token from the developer's own free @BotFather account to activate — the code path is real, but has not been demonstrated against Anthropic's or a judge's own Telegram account, since that requires a locally-run token issued to the operator, not a hosted always-on credential.

4. Detection Methodology & Evaluation

The engine combines two layers rather than relying on either alone:

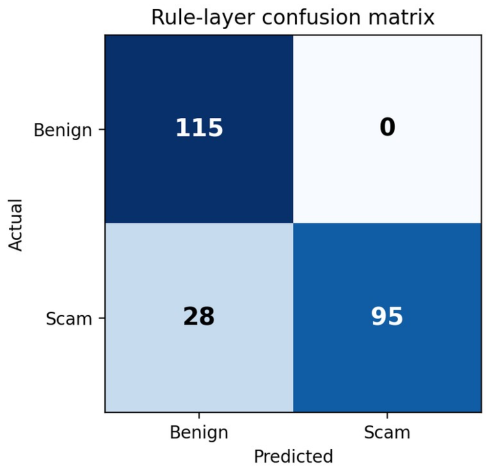
- Rule layer — regex pattern matching across four documented categories of digital-arrest scam mechanics: authority impersonation, forced isolation, urgency/fear, and payment or OTP demand. Fast, explainable, and functions even if the LLM call fails.
- LLM layer — contextual scoring via Mistral 7B (local, through Ollama) for phrasing the rules do not catch, plus a plain-language reason string surfaced directly to the citizen.

The two scores are blended (55% rule-weighted, 45% LLM-weighted). Rules are weighted slightly higher deliberately: false positives on a citizen-facing safety tool carry real cost, and false-positive rate is an explicit judging criterion for this problem statement. Risk is scored on the co-occurrence of categories, not on any single phrase — no individual line is proof of fraud, but authority claim plus isolation instruction plus urgency plus payment demand together is the signature combination scammers actually use.

4.1 Quantitative evaluation

A labeled evaluation set of 238 synthetic, developer-authored transcripts (123 scam, 115 benign — including 38 deliberately adversarial "hard negatives" that use scam-adjacent vocabulary without being scams) was scored against the detector. "Flagged" = medium or high verdict, matching what the UI actually shows a citizen.

Metric	Value (rule layer)
Precision	1.00
Recall	0.77
F1	0.87
False positive rate — overall	0.00
False positive rate — hard negatives	0.00
Confusion matrix	TN 115, FP 0, FN 28, TP 95



These numbers are for the rule layer alone, evaluated in an offline development environment without Ollama running. A held-out spot check outside the generated set (including a Hindi-English code-switched scam script) confirmed the expected limitation: the rule layer misses genuinely novel phrasing, scoring it 0. This is reported deliberately, not hidden — it is exactly the gap the LLM layer exists to close, and precision/FPR (the criteria this problem statement weights most heavily for citizen-facing tools) are already strong on the rule layer alone. Full methodology, the generation script, and instructions to reproduce the hybrid evaluation with Ollama running are in backend/eval/.

5. Fraud Network Graph Intelligence

A second pillar addresses the problem statement's "Fraud Network Graph Intelligence" build area directly: multiple citizen reports are cross-referenced for shared infrastructure — phone numbers, bank account numbers, UPI IDs, and case references — extracted via regex entity extraction. Reports sharing an entity are connected in a graph; connected components of size 2 or more are surfaced as a likely fraud ring, ranked by size.

On a 14-report synthetic demo set (backend/data/fraud_reports.json), the system correctly identifies two distinct fraud rings (5 reports sharing an account, phone number, and case reference; 4 reports sharing a UPI ID) while correctly leaving 5 unrelated singleton reports unclustered. This is exposed via GET /graph/demo and POST /graph/analyze, and rendered as a graph visualisation in the frontend's “Fraud Network” tab.

This is the same “compound intelligence” principle the detection engine uses — no single report proves an organised operation, but shared infrastructure across multiple victims does — applied one level up, from a single transcript to a population of reports.

6. Data & Ethics

No real victim data, leaked scam scripts, or scraped call recordings were used anywhere in this project. All transcripts, evaluation examples, and fraud-report entities are synthetic and developer-authored, reflecting only the publicly documented mechanics of digital-arrest scams (MHA/RBI public advisories), never reproducing real case material.

7. Technology Stack

Every component in this stack is free and open-source. No paid API keys or subscriptions are required to run or demo the project.

Layer	Choice
Backend	Python, FastAPI (open-source)
Detection logic	Regex rule engine + Mistral 7B (Apache-2.0) via Ollama, running locally — free, no API key, no per-call cost
Graph intelligence	NetworkX + Matplotlib (open-source), regex entity extraction
Evaluation	scikit-learn (precision/recall/F1/confusion matrix)
Frontend	React (Vite), open-source
Citizen channel	Telegram Bot API — free, official, no business verification required
Alerting	Free Telegram message to a saved emergency contact

8. Key Features Delivered

- Real-time transcript risk scoring with a visible 0–100 gauge and low/medium/high verdict, backed by a published precision/recall/FPR evaluation.
- In-text highlighting of the exact phrases that triggered each risk category, so the citizen sees why, not just a number.
- One-click guided report draft, pre-formatted for submission to cybercrime.gov.in or a local police station.
- A real inbound Telegram bot (not a stub): forward a message, get a live verdict back.
- Emergency-contact alert sent as a real, free Telegram message (falls back to a clearly-labeled simulation if not configured).
- Fraud network graph intelligence: detects fraud rings across multiple citizen reports via shared-infrastructure clustering, with a rendered visualisation.

9. Business Impact

Digital arrest scams cost Indian citizens over ₹1,776 crore in nine months of 2024 alone. Scam Shield targets the earliest possible intervention point — the moment a citizen or a family member suspects something is wrong — rather than the point of financial loss or complaint filing, where nearly all existing tooling operates today. The graph intelligence layer additionally gives law enforcement a way to prioritise: a 5-victim ring sharing one account is a materially different response than five unrelated one-off complaints.

The detection engine is designed to generalise beyond a citizen-facing app: it can sit behind a telecom carrier's call-screening pipeline, a bank's fraud desk, or NCRB's citizen reporting channel as a scoring API, without any change to the core logic.

10. Scalability

- Stateless FastAPI service — horizontally scalable behind a load balancer with no architectural change.
- Rule layer requires no inference cost and can run at telecom scale; LLM layer can be sampled or cached for high-volume deployments.
- Graph intelligence scales sub-linearly per new report since entity extraction is independent per report; clustering re-runs incrementally on the existing graph rather than from scratch.
- Multi-language support is a scoped next step, not yet built (see Section 12).

11. Judging Criteria Alignment

Criterion	How this project addresses it
Innovation (25%)	Explainable hybrid scoring (rules + LLM) with a visible reason string; compound-risk graph clustering across reports, not just single-message scoring
Business Impact (25%)	Directly targets a ₹1,776 crore/year fraud category at its earliest intervention point; graph layer supports enforcement prioritisation
Technical Excellence (20%)	Published precision/recall/F1/FPR against a labeled evaluation set with adversarial hard negatives, not an unmeasured demo
Scalability (15%)	Stateless API architecture, ready for carrier/bank/NCRB integration
User Experience (15%)	Real Telegram bot citizen channel; one-tap report

	generation and alerting

12. Deliverables & Honest Status

Deliverable	Status
Working prototype	Done — detection engine, graph intelligence, web UI, inbound Telegram bot, all runnable locally with zero paid dependencies
Architecture diagram	Done — Section 3
Quantitative evaluation	Done — Section 4.1, reproducible via backend/eval/
Presentation deck	In progress — to accompany this submission
Demo video (3–4 min)	Scripted, pending recording — shot list available on request
Regional-language support	Not yet built — scoped as the next priority (Section 9 of README roadmap)

13. Repository & Demo

GitHub repository: [add link after pushing]

Live demo / video: [add link]